METHODIST
COLLEGE OF ENGINEERING & TECHNOLOGY
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AI and Blockchain Based eKYC Verification System

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Abstract

In today's digital world, verifying identity accurately and securely is critical.Traditional KYC methods are time-consuming, centralized, and vulnerable to data breaches.This project introduces an AI-driven eKYC system integrated with Blockchain to achieve fast, tamper-proof, and privacy-focused identity verification.Artificial Intelligence is used to extract and validate user data through facial recognition and document OCR, ensuring authenticity and eliminating manual intervention.The Blockchain layer records only encrypted hashes of verified identities, providing transparency and immutability while keeping personal data private.

Introduction

- In today's digital era, verifying the identity of individuals securely and efficiently has become a fundamental requirement for banks, telecom providers, and government platforms. Traditional KYC processes are manual, time-consuming, and prone to data misuse, often exposing sensitive user information.
- eKYC (Electronic Know Your Customer) provides a digital, paperless, and automated method for identity verification. It reduces verification time from days to seconds, improving both user experience and operational efficiency.
- However, with growing digitalization, privacy and data security have become major concerns. Sensitive personal details such as Aadhaar numbers, biometric data, and photos must be protected from unauthorized access or alteration.
- This project addresses these challenges by integrating Artificial Intelligence for intelligent identity validation and Blockchain for decentralized, tamper-proof, and privacy-preserving data storage — ensuring that user information remains secure, private, and verifiable across multiple platforms.

Literature Survey

Topic	Paper	Summary	Strengths	Weaknesses	Relevance/Gaps
Biometric + Blockchain Identity Systems	Blockchain-based biometric identity management (Cluster Computing / Springer)	Manages face biometric template hashes on private blockchain; benchmarking on common datasets.	Practical demo of template integrity using blockchain.	Limited anti-spoofing; selective disclosure not fully addressed.	Validates storing biometric commitments on-chain; gaps in privacy and liveness checks.
Privacy-Preserving Biometric Authentication on Public Chains	BioZero (arXiv)	Uses cryptographic commitments, homomorphic operations & zero-knowledge proofs for privacy.	Strong privacy guarantees suitable for public blockchain.	High computational cost; complex crypto stack required.	Blueprint for privacy; suitable for future ZK upgrades; current focus on pragmatic Merkle-tree approach.
Document Field Extraction + Merkle-Based Selective Disclosure	Blockchain-Based Decentralized Identity Management (MDPI / Computers)	AI extracts document fields; commits Merkle root on-chain for selective proof.	Practical selective disclosure & minimized on-chain storage.	OCR may lack robustness under varied conditions; gas costs a concern.	Approach directly supports selective disclosure & privacy for KYC documents.
Non-Disclosing On-Chain Credentials & Verifiable Claims	Non-Disclosing Credential On-chaining (arXiv)	Issues verifiable credentials with privacy via zero-knowledge & selective disclosure.	Standards-oriented; interoperable with DID/VC frameworks.	Needs extra infrastructure; complexity for small demos.	Useful for future extensibility to VC standards and revocation.
Surveys and Systematic Reviews	SoK: Bridging Trust into the Blockchain (arXiv)	Reviews on-chain identity across blockchains, DIDs, ZK proofs, and biometrics.	Highlights hybrid designs balancing privacy, cost & usability.	None specifically.	Reinforces hybrid off-chain/on-chain design as practical architecture.

Literature Survey

S. No.	Paper & Authors (Year)	Core Idea	Techniques Used	Limitations	Relevance to Proposed Work
1 (Base)	Piper et al. (2025)	Privacy-preserving on-chain permissioning enabling decentralized access control with anonymous credentials	ZK-SNARKs, SSI, ABAC	Focused mainly on regulatory compliance; lacks abuse-reporting workflow	Forms architectural backbone for anonymous reporting using ZKP verification
2	Sharma et al. (2025)	Secure private reads on permissioned blockchain to protect sensitive data queries	Homomorphic Encryption, PIR	Limited to Fabric network; computationally heavy for real-time systems	Supports design of privacy-preserving threat-score lookup mechanism
3	Riaz et al. (2025)	Confidential trust validation combining blockchain proofs with trusted hardware environments	ZKP + Trusted Execution Environments	High computation overhead and hardware dependency	Inspires secure off-chain scoring and governance logic
4	Bose et al. (2025)	Hybrid blockchain–cloud architecture for encrypted data access and secure collaboration	ZKP, Homomorphic Encryption	Mostly conceptual; limited prototype validation	Helps structure encrypted evidence submission and verification pipeline

Literature Survey

S. No.	Paper & Authors (Year)	Core Idea	Techniques Used	Limitations	Relevance to Proposed Work
5	Khalid et al. (2025)	Privacy-preserving digital signatures enabling anonymous authentication across distributed networks	Blind, Ring, Aggregate Signatures	Does not include reporting or governance mechanisms	Useful for designing privacy-friendly identity verification and anonymous proof models
6	Sohani et al. (2025)	Blockchain-based decentralized reputation framework for evaluating trustworthiness in networks	DAG-based consensus, Reputation Scoring	Missing anonymity features and privacy safeguards	Direct foundation for threat-score and decentralized trust computation
7	Khan et al. (2025)	Secure CI/CD pipelines using blockchain audit logs for transparent monitoring	Hyperledger Fabric, Audit Trails	DevOps-specific focus; not tailored for identity or reporting	Provides backend auditability and tamper-proof logging inspiration
8	Patel et al. (2024)	Hybrid security architecture combining anomaly detection with blockchain-cloud infrastructure	Machine Learning Anomaly Detection, Hybrid Architecture	Limited focus on privacy-preserving identity systems	Contributes ideas for threat signal detection and system monitoring modules

Motivation

Rising Identity Fraud & Data Breaches

•Digital identity fraud cases are increasing due to fake documents, spoofing, and deepfake attacks.
•Centralized KYC databases create single points of failure and hacking risks .
•Manual verification leads to delays, human errors, and high operational cost.

Rapid Growth of Digital Onboarding

•Millions of users onboard monthly across banking, telecom, fintech, and e-commerce platforms.
•Traditional KYC processes cannot scale with modern digital transformation .
•Users demand **instant verification** and seamless onboarding experiences.

Motivation

Repeated KYC Problem

One user must submit identity verification repeatedly across multiple services.

Causes:

- Redundant effort
- Increased cost
- Poor user experience

Blockchain enables **one-time verification with reusable identity proof**.

Privacy & Trust Challenges

- Storing raw identity data increases risk of breaches.
- AI alone improves accuracy but cannot ensure tamper-proof records.
- Blockchain ensures immutable verification history without exposing sensitive data

Statistical Data

Growth of Digital Identity & eKYC

•Over **1.3 billion Aadhaar IDs** have been issued in India, making it the world's largest digital identity system.

•More than **90% of bank accounts** in India are linked with Aadhaar-based eKYC.

•Digital onboarding using eKYC reduces customer onboarding time from **days to minutes**.

Identity Fraud & Data Breaches

•Identity fraud contributes to **30–40% of financial fraud cases** globally.

•Centralized identity databases are prime targets for cyberattacks and data leaks.

•Global data breaches expose **billions of identity records every year**, increasing misuse risks.

Statistical Data

Cost & Efficiency of eKYC

- Traditional KYC costs financial institutions 5–10 times more than digital eKYC.
- Repeated KYC across banks, telecoms, and fintech platforms leads to significant redundancy and cost overhead.
- Blockchain-based verification can enable one-time KYC with reusable identity proof.

Technology Adoption Trends

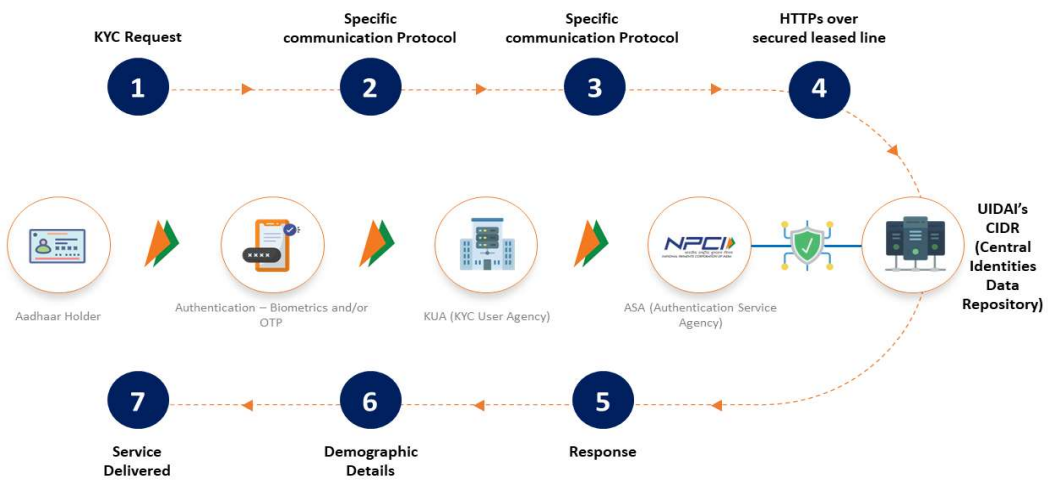
- AI-based biometric verification achieves 95–99% accuracy in controlled environments.
- Blockchain adoption in identity management is growing due to its immutability and auditability.
- Privacy-preserving technologies (hashing, ZKPs) are increasingly used to comply with data protection regulations

Problem Statement

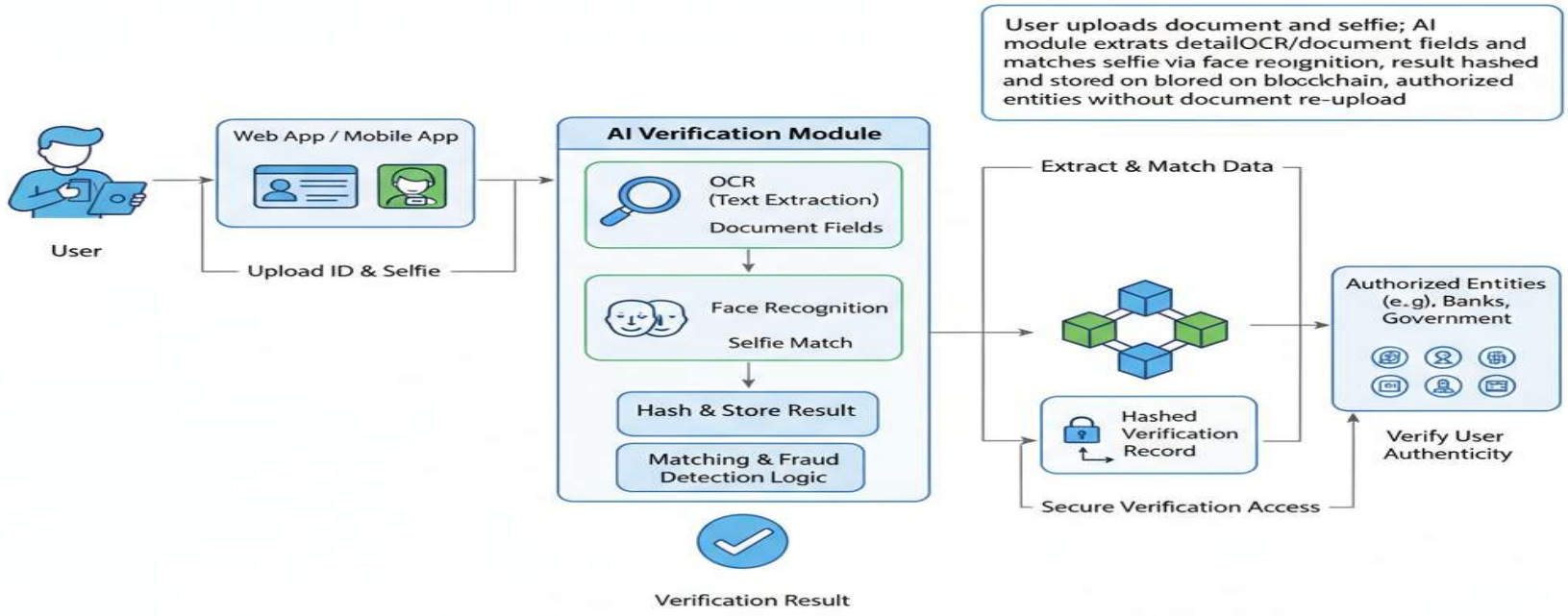
Traditional KYC systems are slow, manual, and prone to data breaches. Centralized storage of sensitive identity information increases the risk of misuse. Users repeatedly submit the same documents to different institutions, increasing risks of data theft, forgery, and privacy violations. There is a need for a decentralized, AI-driven solution that ensures both efficiency and security.

Objectives

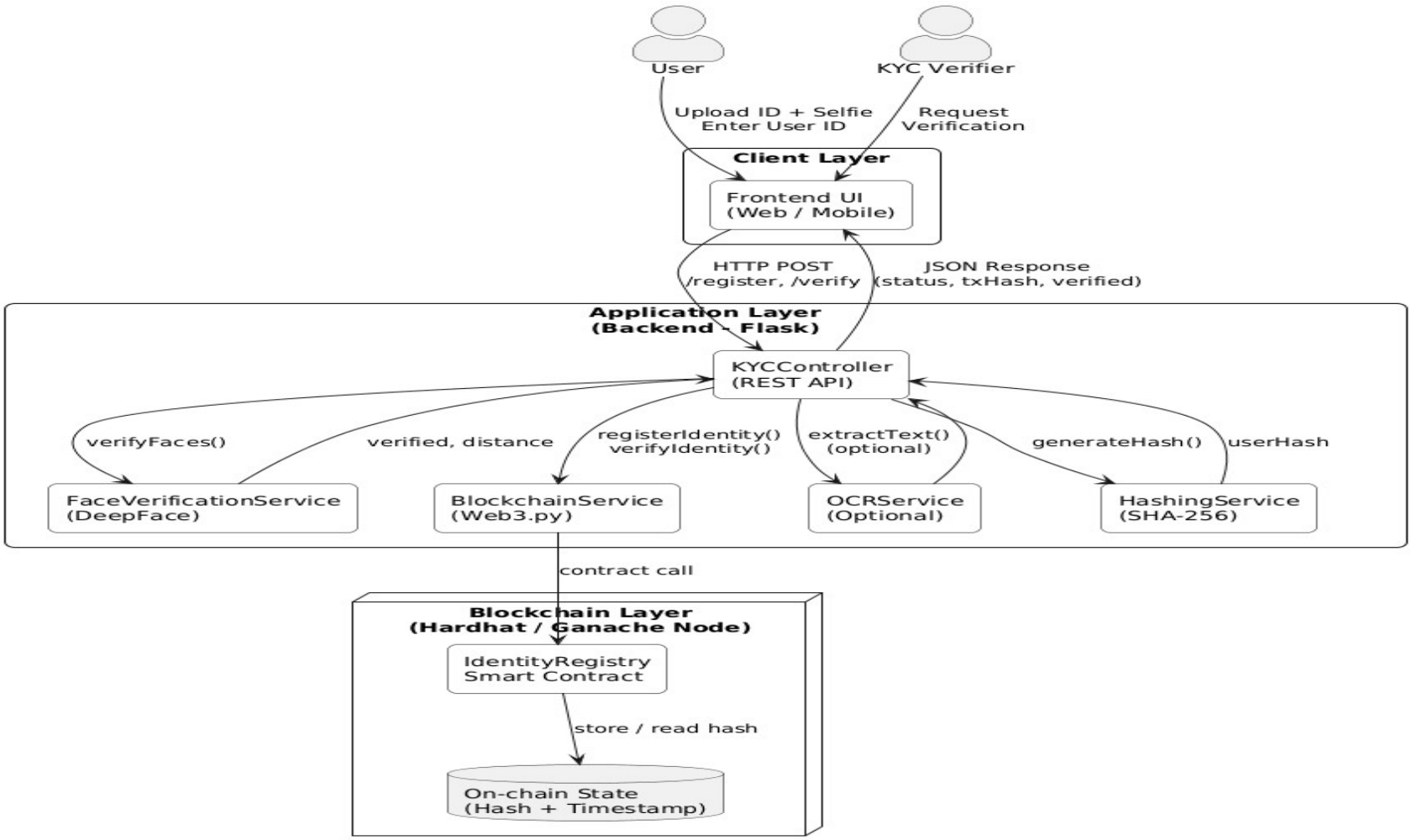
- To automate identity verification using AI-based OCR and face recognition.
- To securely store verification proofs using blockchain.
- To ensure privacy through encrypted and decentralized data storage.
- To eliminate redundant KYC processes across organizations.



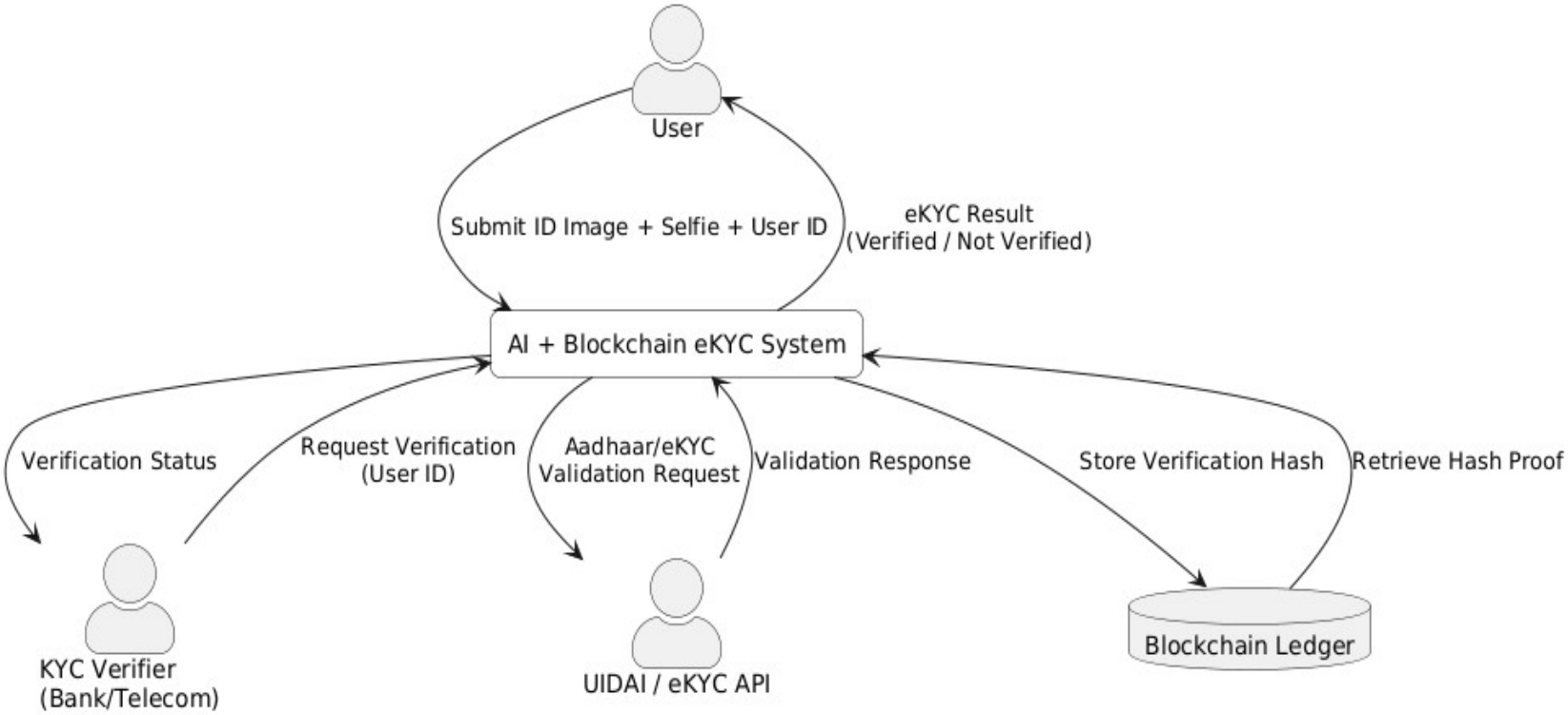
Proposed Architecture



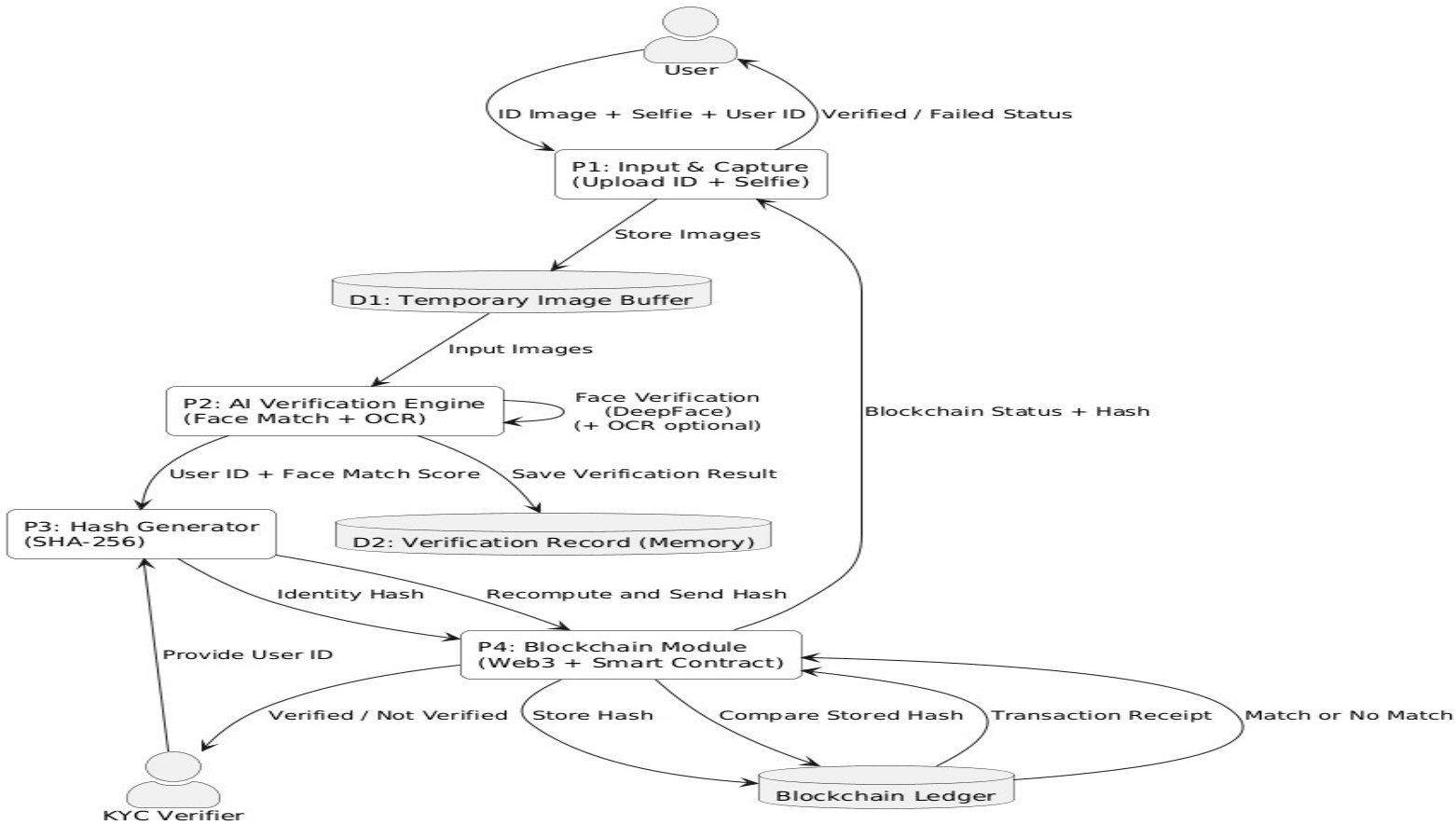
Final Architecture



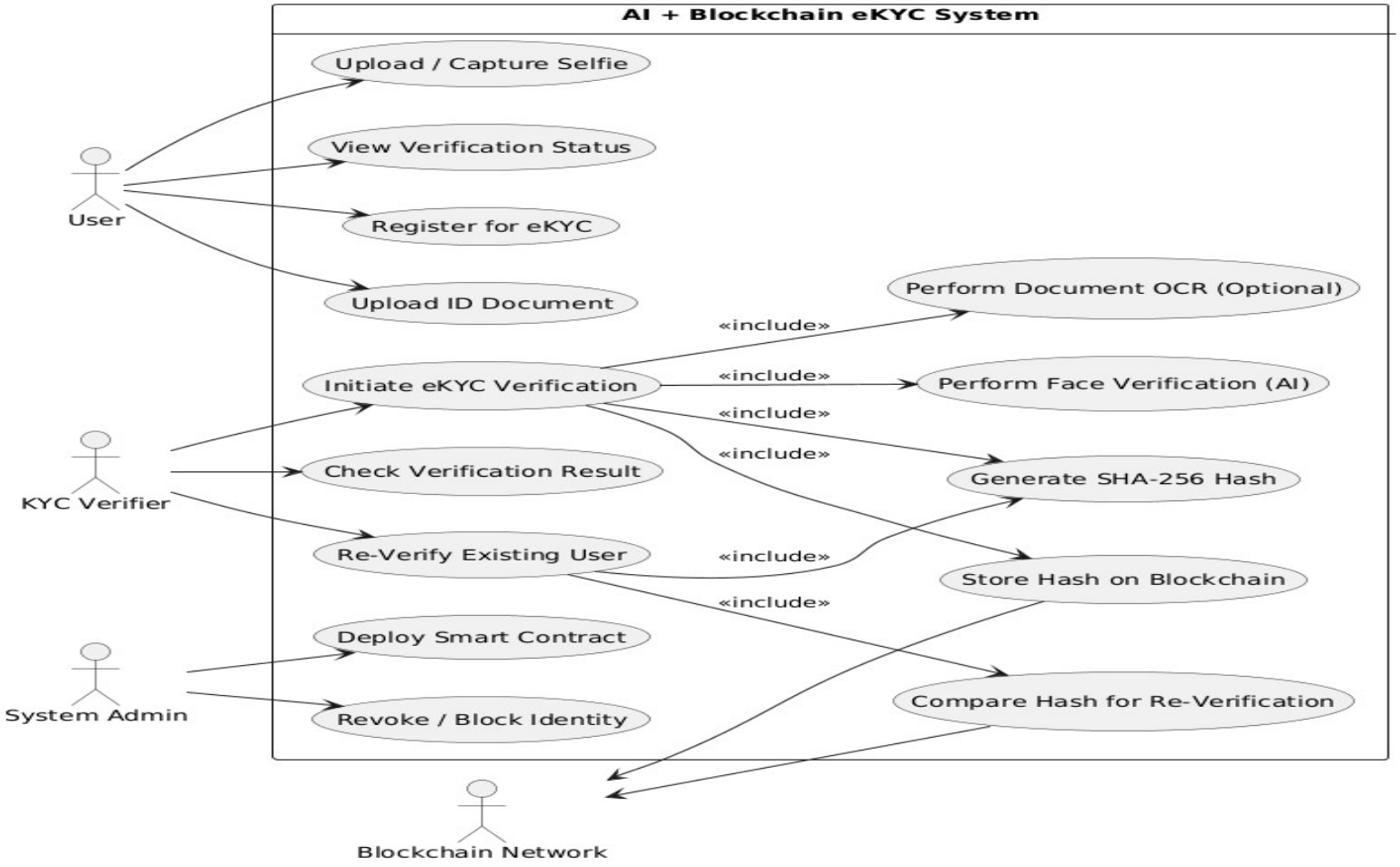
Data Flow Diagram-Level-0



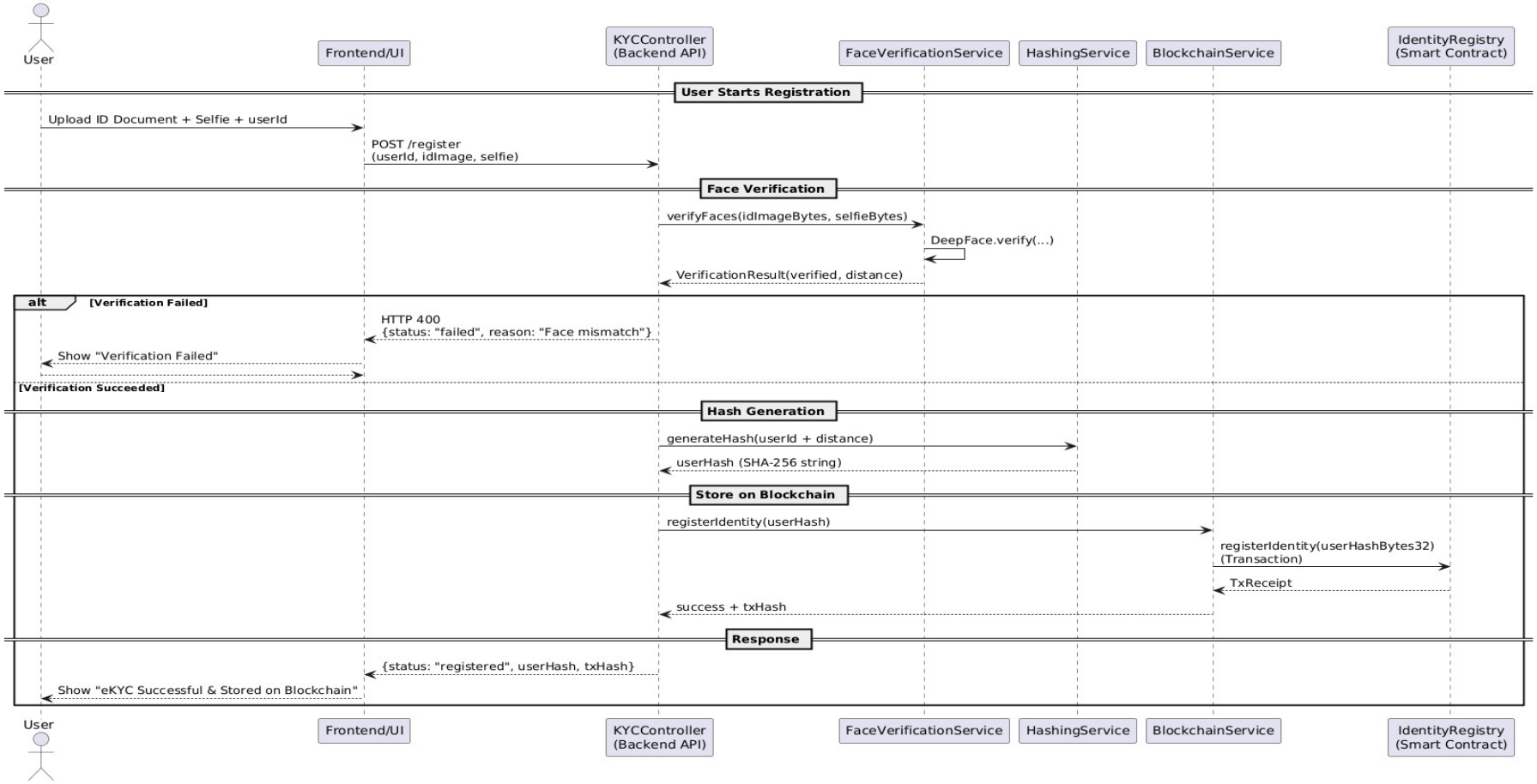
Data Flow Diagram-Level-1



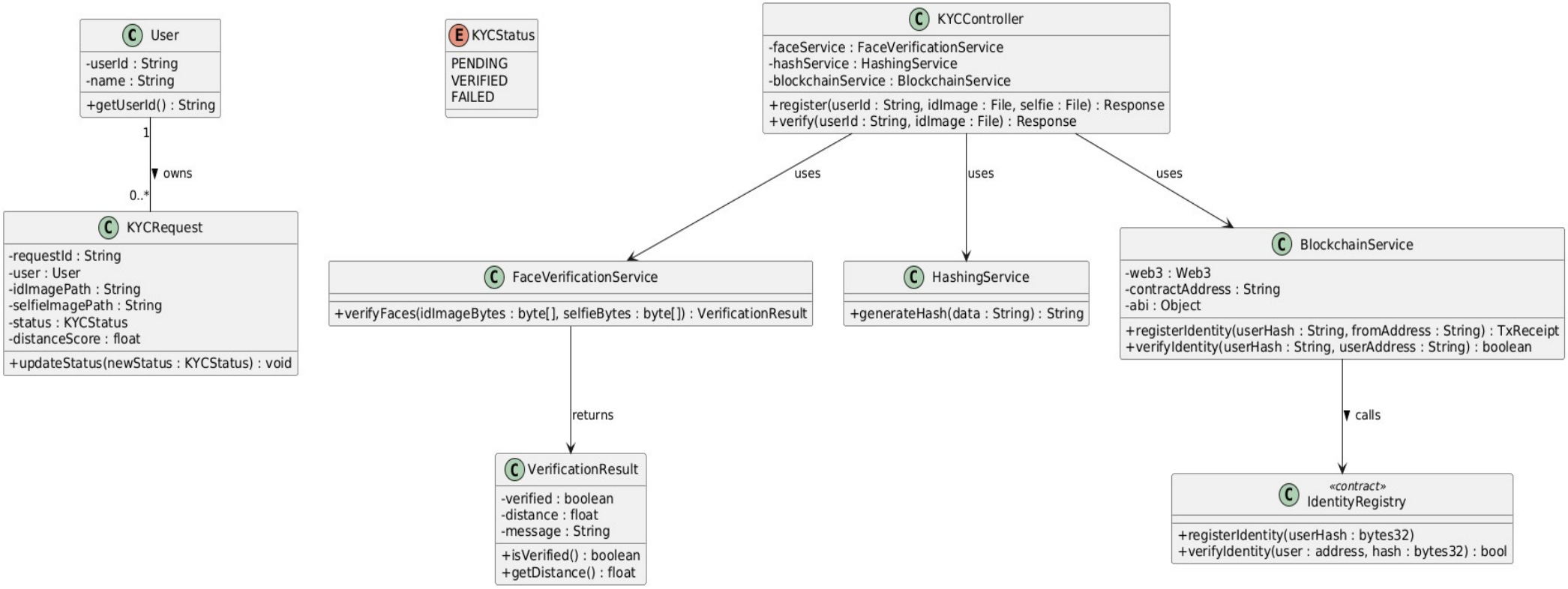
UseCase Diagram



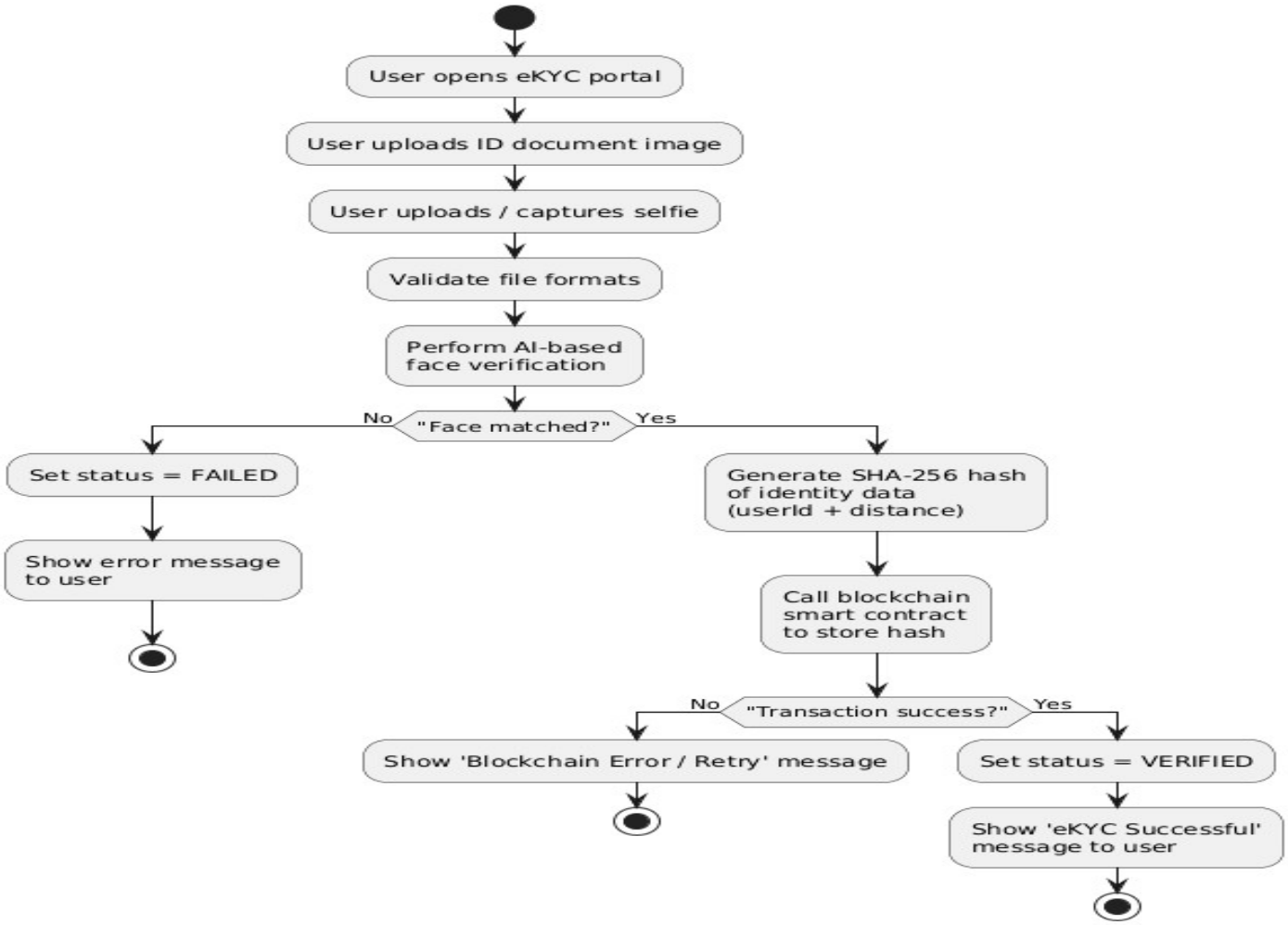
Sequence Diagram



Class Diagram



Activity Diagram



Algorithm

Step 1: START

Step 2: Accept ID_Image, Selfie_Image, and User_ID from application interface.

Step 3: Perform preprocessing
- Validate file format
- Resize images
- Normalize input data

Step 4: Extract document fields using OCR module
Fields ← OCR(ID_Image)

Step 5: Perform face verification
Match_Result ← DeepFace_Verify(ID_Image, Selfie_Image)
Similarity_Score ← Match_Result.score

Algorithm

Step 6: Check verification threshold

```
IF Similarity_Score < Threshold THEN
    Verification_Status ← "Rejected"
    RETURN Verification_Status
ENDIF
```

Step 7: Execute fraud detection logic

- Check abnormal scores
- Validate extracted fields

Step 8: Generate secure hash

```
Hash_Input ← concatenate(User_ID, Similarity_Score, Verification_Status)
User_Hash ← SHA256(Hash_Input)
```

Step 9: Store verification proof on blockchain

```
Transaction_ID ← SmartContract.store(User_Hash)
```

Step 10: Provide verification access

Authorized entities verify using User_Hash

Step 11:RETURN Verification_StatusandTransaction_ID

Step 12: Stop

Implementation-System Design

Development Environment

•Programming Language: Python
•Framework: Flask (Backend API)
•AI Models: DeepFace (Face Verification), OCR Engine
•Blockchain: Ethereum (Hardhat/Ganache)
•Smart Contract: IdentityRegistry
•Frontend: Web-based Interface

Architecture Integration

•User Interface connects to backend through REST APIs
•AI modules perform verification
•Blockchain stores hashed verification proof

Implementation-Analysis

Data Processing Pipeline

- Image preprocessing (resize, normalization)
- OCR extraction of document fields
- Face similarity computation using DeepFace

Security Analysis

- SHA-256 hashing applied to verification results
- No personal identity data stored on blockchain
- Ensures privacy-preserving verification

Performance Considerations

- Local blockchain reduces transaction cost
- AI inference optimized using pretrained models

Implementation-Modules

User Interface Module

- Upload ID document and selfie
- Displays verification status

AI Verification Module

- OCR-based text extraction
- Face recognition & similarity scoring
- Fraud detection logic

Blockchain Module

- Smart contract deployment
- Store verification hash
- Retrieve verification status

Implementation-Workflow

System Execution Steps

- User uploads identity document and selfie.
- OCR extracts document information.
- DeepFace performs face matching.
- Verification result converted into secure hash.
- Hash stored on blockchain smart contract.
- Authorized entity verifies user through blockchain proof.

Implementation-Prediction/Output

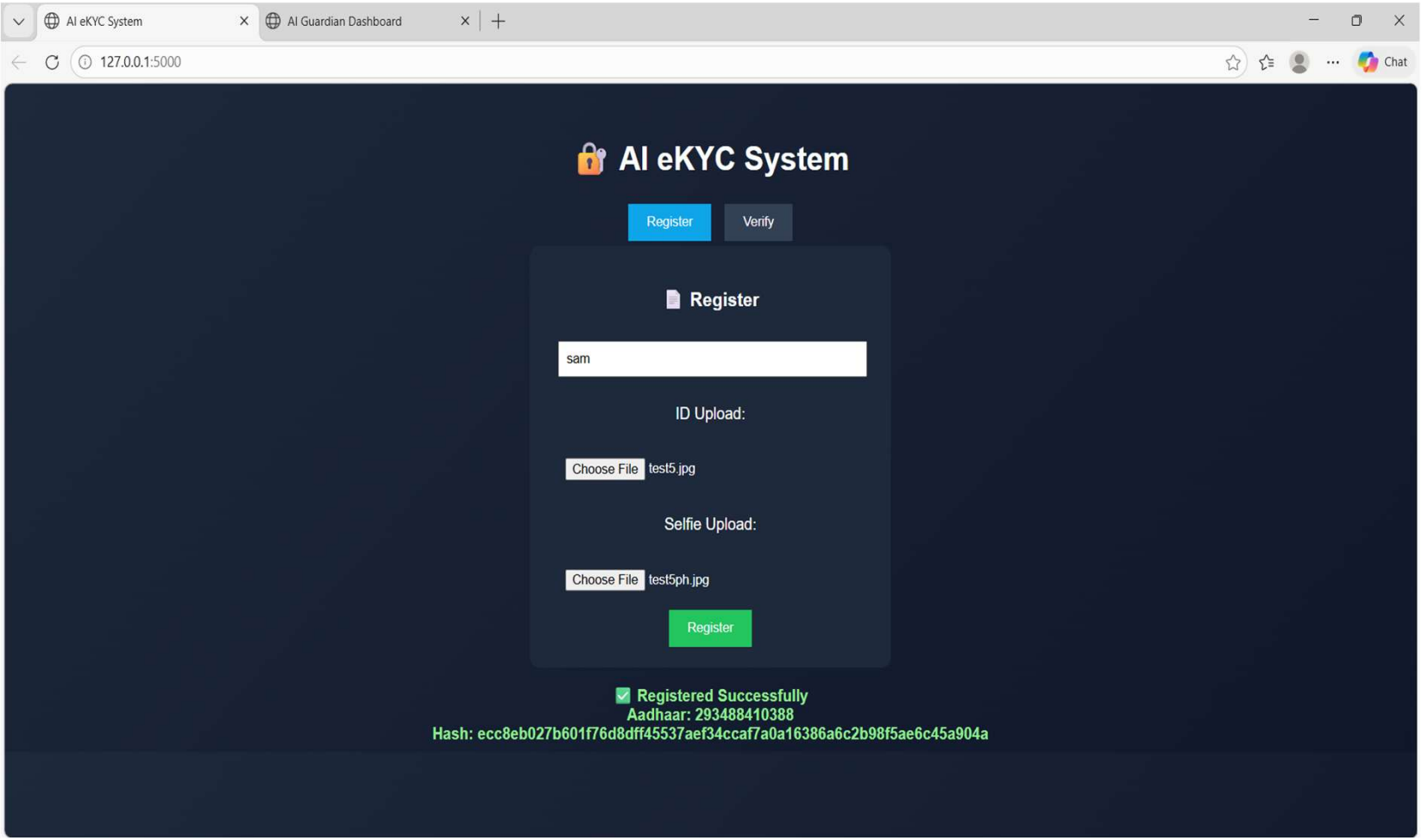
Verification Output

- Verified / Rejected status
- Similarity score
- Blockchain transaction ID

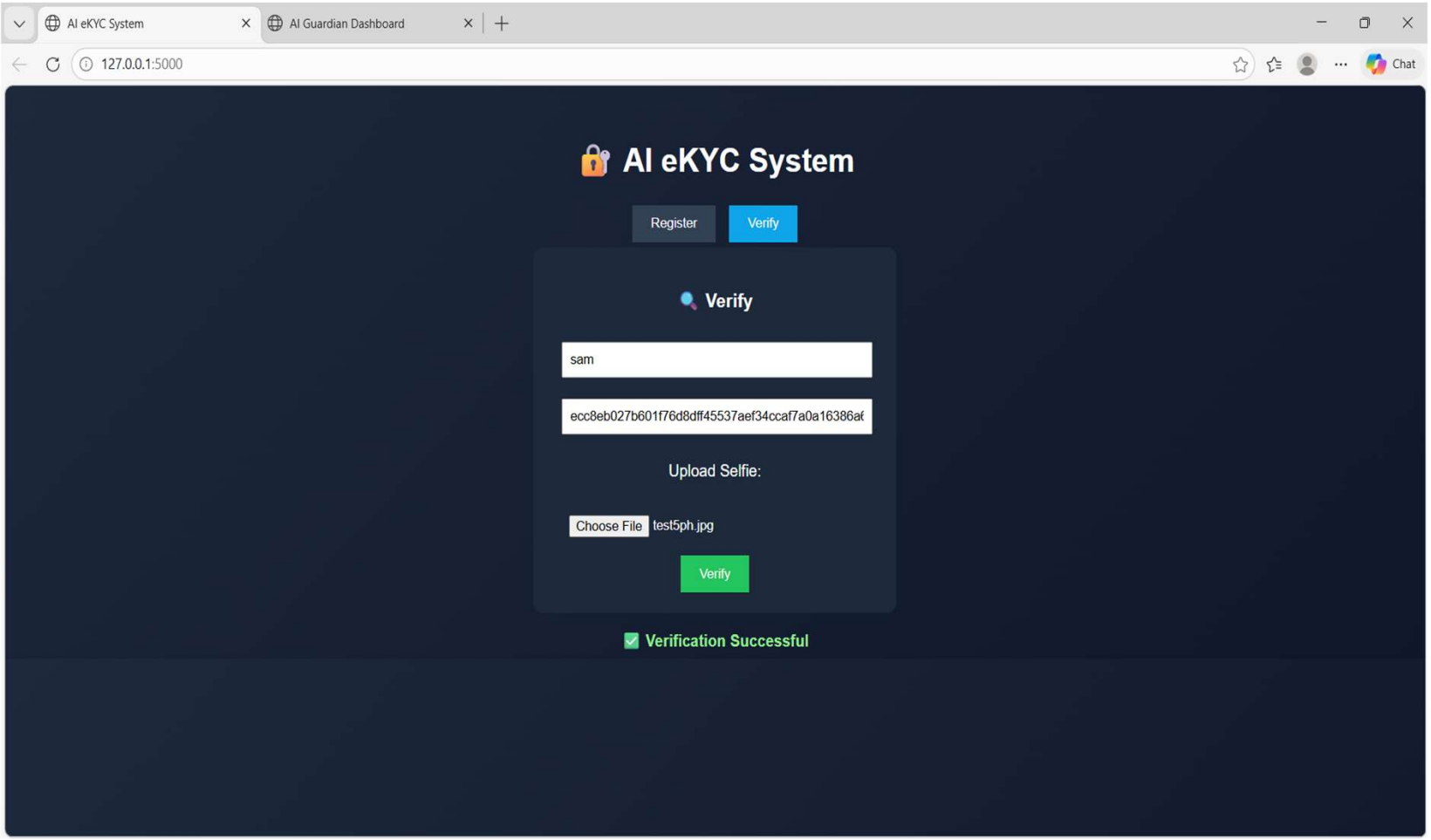
System Advantages

- Eliminates repeated KYC
- Privacy-preserving identity verification
- Tamper-proof blockchain record

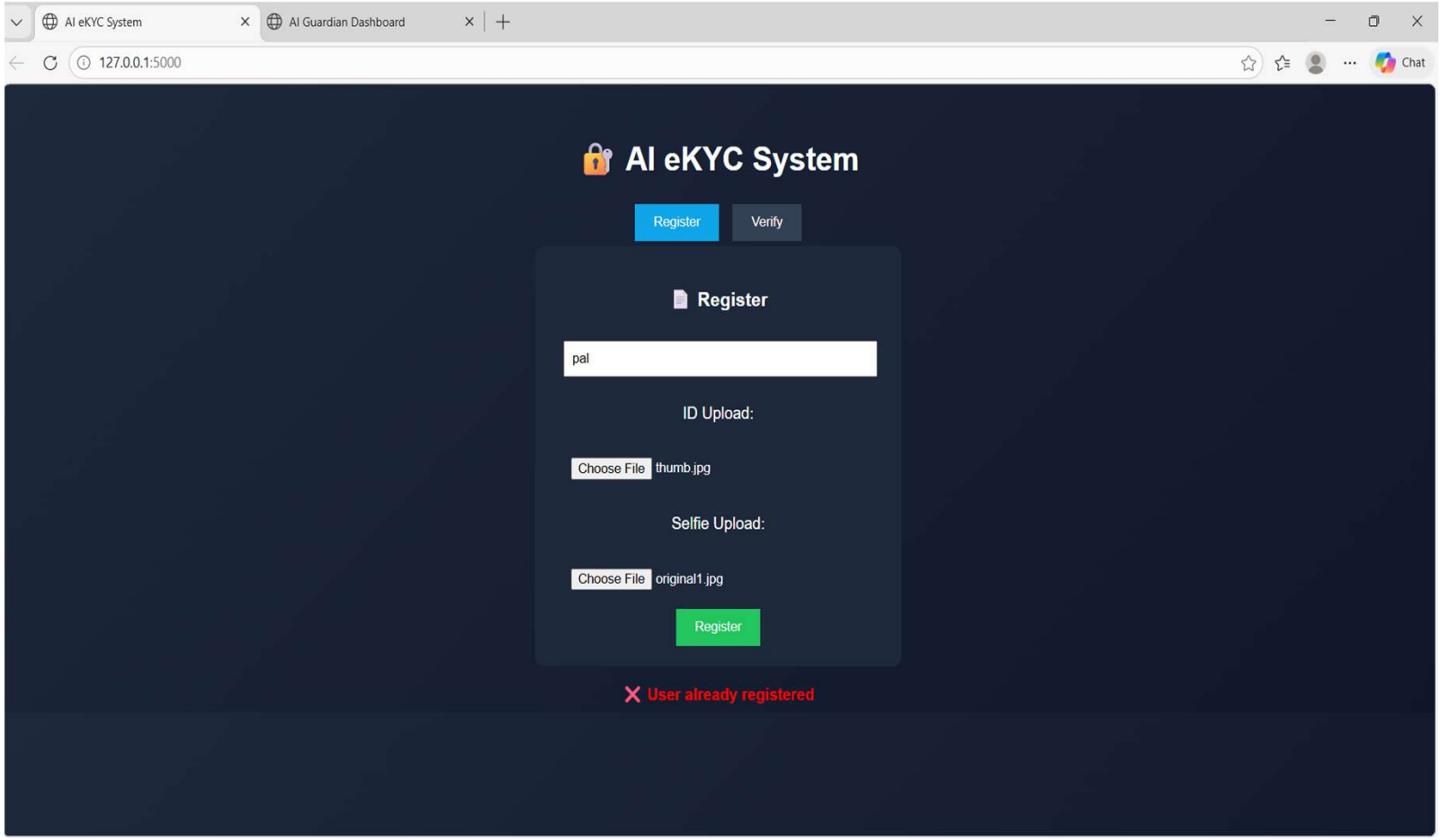
Result


AI eKYC System
AI Guardian Dashboard
127.0.0.1:5000
Chat
AI eKYC System
Register
Verify
Register
sam
ID Upload:
Choose File
test5.jpg
Selfie Upload:
Choose File
test5ph.jpg
Register
Registered Successfully
Aadhaar: 293488410388
Hash: ecc8eb027b601f76d8dff45537aef34ccaf7a0a16386a6c2b98f5ae6c45a904a

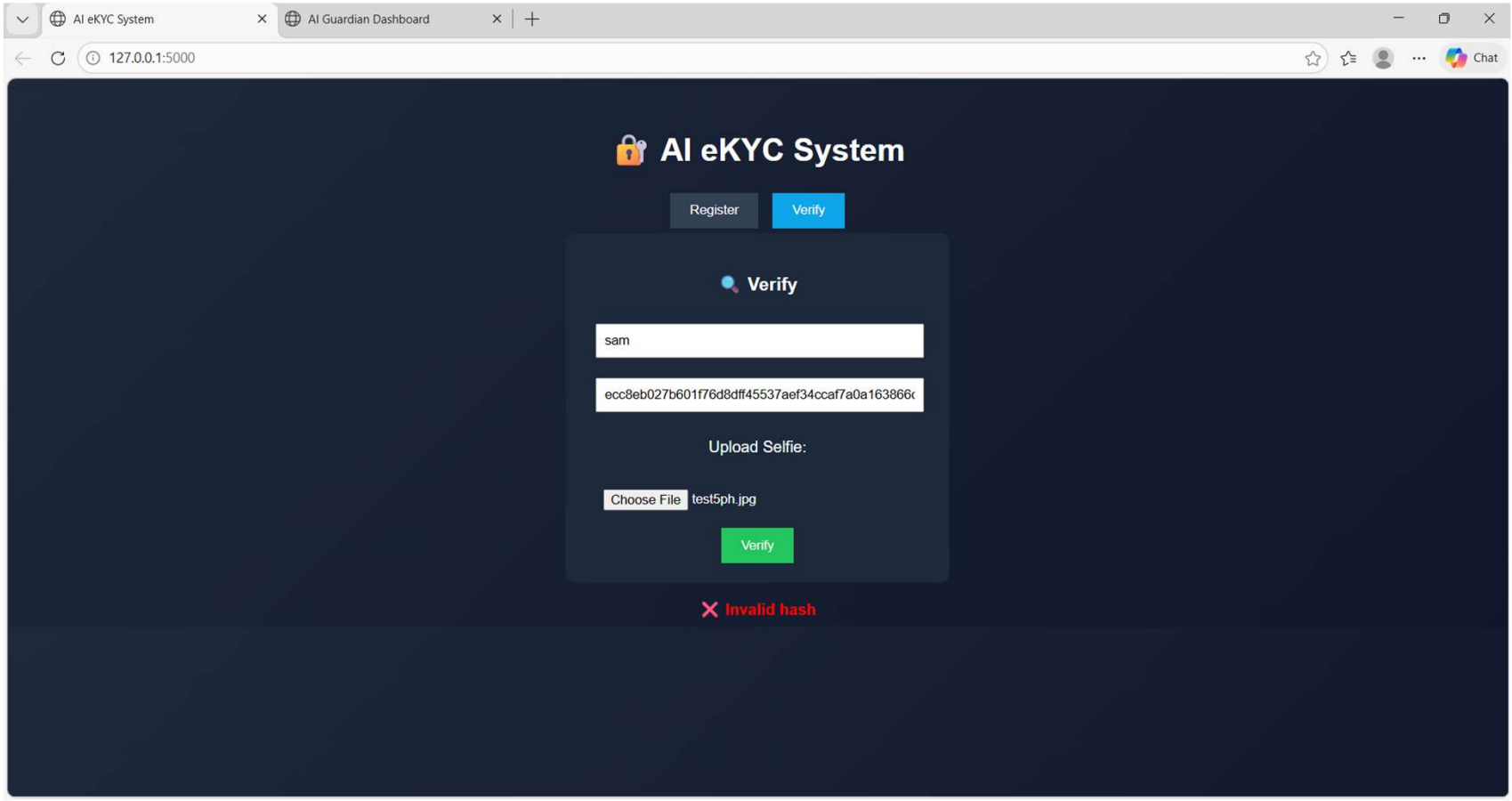
Result


AI eKYC System
AI Guardian Dashboard
127.0.0.1:5000
Chat
AI eKYC System
Register
Verify
Verify
sam
ecc8eb027b601f76d8dff45537aef34ccaf7a0a16386a
Upload Selfie:
Choose File
test5ph.jpg
Verify
Verification Successful

Result


AI eKYC System
AI Guardian Dashboard
127.0.0.1:5000
Chat
AI eKYC System
Register
Verify
Register
pal
ID Upload:
Choose File thumb.jpg
Selfie Upload:
Choose File original1.jpg
Register
User already registered

Result


AI eKYC System
AI Guardian Dashboard
127.0.0.1:5000
Chat
AI eKYC System
Register
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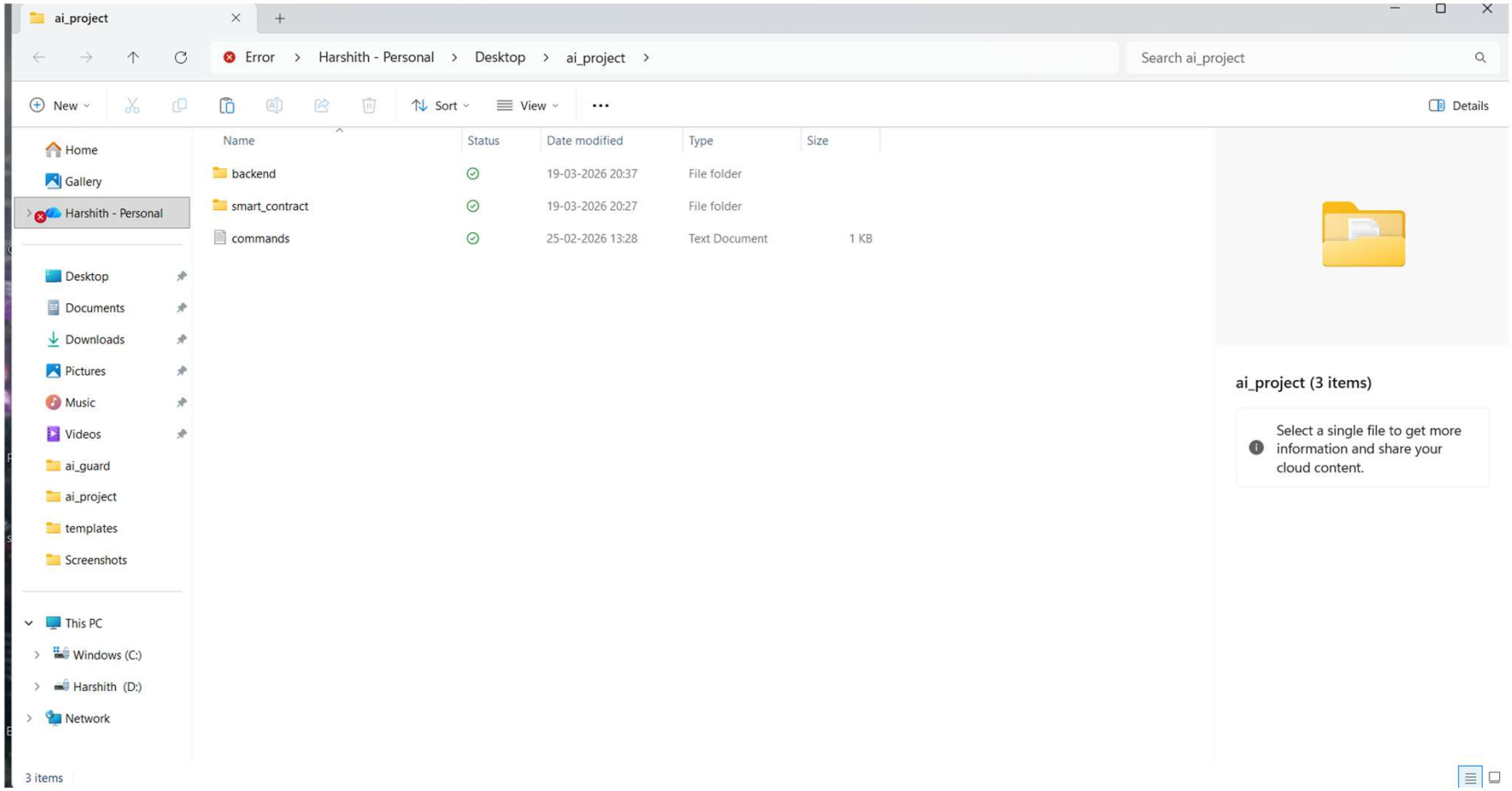
Upload Selfie:
Choose File
test5ph.jpg
Verify
Invalid hash

Result


ai_project
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Network
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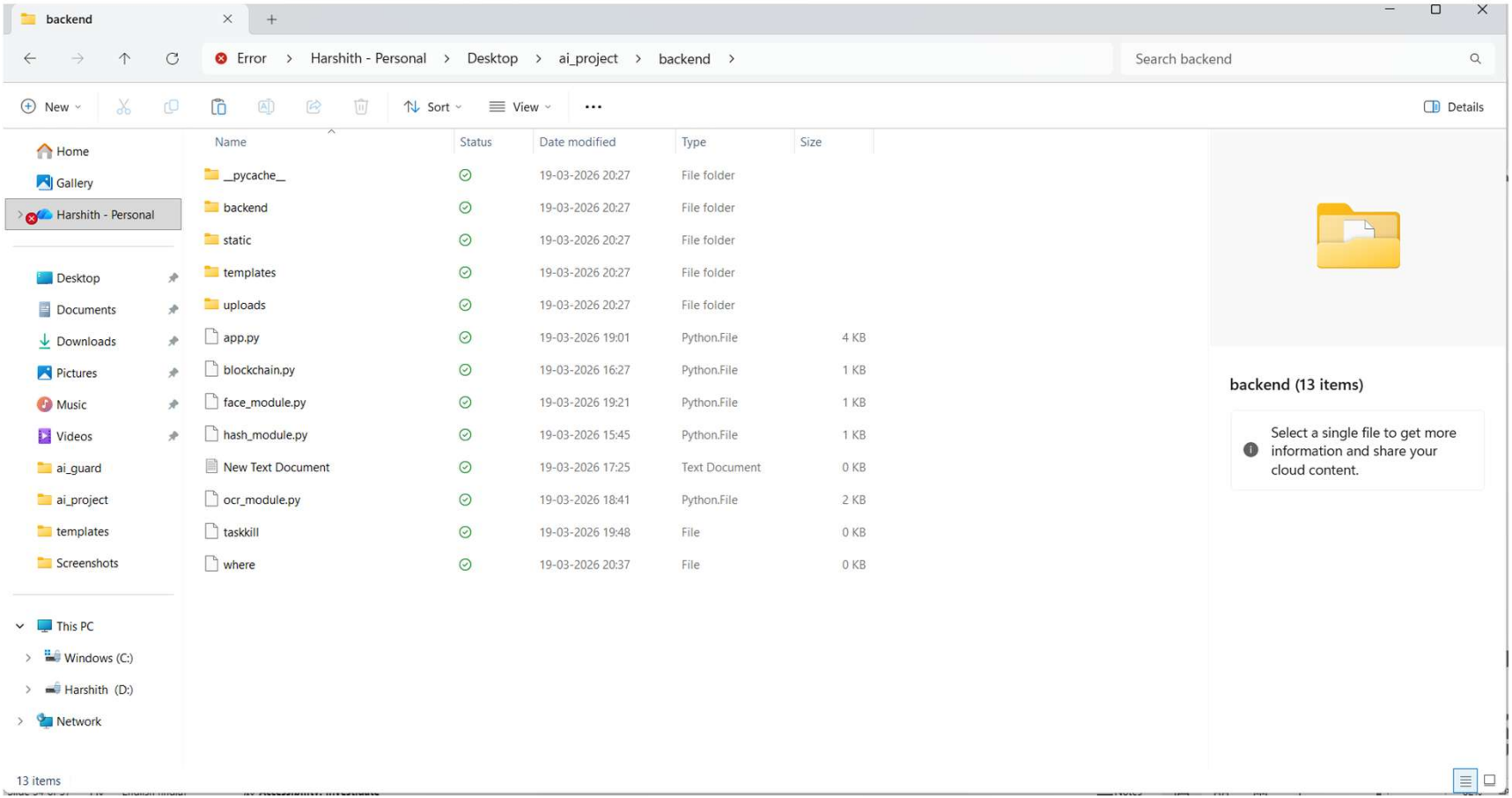
backend
19-03-2026 20:37
File folder
smart_contract
19-03-2026 20:27
File folder
commands
25-02-2026 13:28
Text Document
1 KB
ai_project (3 items)
Select a single file to get more information and share your cloud content.
3 items

Result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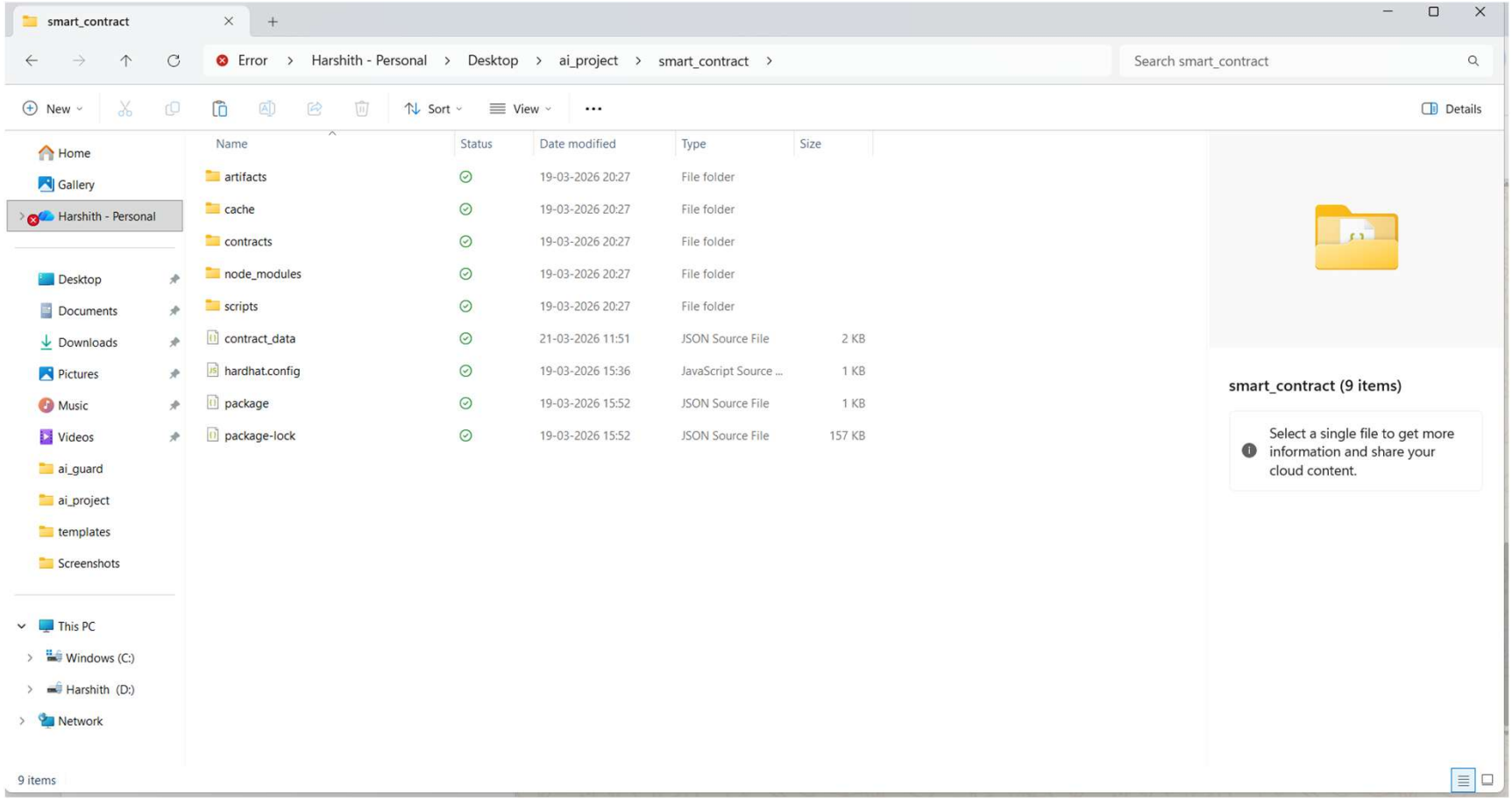
backend 19-03-2026 20:27 File folder
static 19-03-2026 20:27 File folder
templates 19-03-2026 20:27 File folder
uploads 19-03-2026 20:27 File folder
app.py 19-03-2026 19:01 Python.File 4 KB
blockchain.py 19-03-2026 16:27 Python.File 1 KB
face_module.py 19-03-2026 19:21 Python.File 1 KB
hash_module.py 19-03-2026 15:45 Python.File 1 KB
New Text Document 19-03-2026 17:25 Text Document 0 KB
ocr_module.py 19-03-2026 18:41 Python.File 2 KB
taskkill 19-03-2026 19:48 File 0 KB
where 19-03-2026 20:37 File 0 KB
backend (13 items)
Select a single file to get more information and share your cloud content.
13 items

Result


smart_contract
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Home
Gallery
Harshith - Personal
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Documents
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Pictures
Music
Videos
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templates
Screenshots
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Windows (C:)
Harshith (D:)
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Status
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cache 19-03-2026 20:27 File folder
contracts 19-03-2026 20:27 File folder
node_modules 19-03-2026 20:27 File folder
scripts 19-03-2026 20:27 File folder
contract_data 21-03-2026 11:51 JSON Source File 2 KB
hardhat.config 19-03-2026 15:36 JavaScript Source ... 1 KB
package 19-03-2026 15:52 JSON Source File 1 KB
package-lock 19-03-2026 15:52 JSON Source File 157 KB
smart_contract (9 items)
Select a single file to get more information and share your cloud content.
9 items

Limitations Of the System

Drawbacks of Blockchain for eKYC:

- - Strict UIDAI and RBI rules restrict storing Aadhaar or personal data externally.
- - Blockchain immutability conflicts with data deletion laws (DPDP 2023).
- - Integration with existing centralized KYC systems is complex.
- - Lack of blockchain awareness and regulatory framework in India.

Scalability of the Project:

- - Supports large-scale multi-user verification through distributed blockchain nodes.
- - Can integrate with multiple sectors (banking, telecom, healthcare, e-governance).
- - Uses pre-trained AI models capable of handling millions of verifications efficiently.
- - Can expand from local blockchain to national or global decentralized identity networks.

Timeline

Phases	Key Tasks	Deliverables / Milestones
Phase 1	Planning, architecture design, environment setup, UI wireframes	Finalized system architecture, project setup complete
Phase 2	User registration + upload; AI OCR & face recognition integration	Working upload + AI modules
Phase 3	Blockchain integration, linking AI outputs to chain, selective disclosure, access control	Blockchain module live, verification workflow tested
Phase 4	Security (liveness, anti-spoofing), end-to-end testing, deployment	Secure MVP deployed, user testing completed

Conclusion

The AI and Blockchain-based eKYC system provides a secure, transparent, and efficient method for digital identity verification. It eliminates central vulnerabilities, prevents fraud, and ensures user privacy. This project demonstrates a scalable and practical solution for the future of decentralized identity management. By using this services like online passport verification and online voting can also be done